TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Abrasion Resistance of Fabric

The Martindale test: a Lissajous rub against a standard abradant, an endpoint counted in thousands of rubs, and a mass-loss check that watches the fabric wear away.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 7 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the abrasion resistance of the reference fabric on the Martindale apparatus — by the thread-breakdown endpoint (rubs to the breaking of two threads) and by the mass-loss method at a fixed number of rubs — and to note the related surface failure, pilling.

1.1 The Martindale principle

A circular specimen, held in a holder under a defined pressure, is rubbed against a standard abradant (a specified worsted fabric mounted on the abrasion table, backed by felt) while the holder is driven in a **Lissajous motion** — a pattern that slowly rotates through every direction, tracing curves that continuously change orientation. The Lissajous figure is the point of the instrument: real abrasion in wear attacks from shifting directions, and a simple back-and-forth rub would wear a directional groove that flatters some weaves and punishes others. One traversal of the complete Lissajous figure comprises 16 rubs, and the counter total is reported simply as **rubs** (cycles).

The nominal pressure is set by the holder mass and mounting: **9 kPa for apparel fabrics** and 12 kPa for workwear and upholstery-type end uses. This sheet uses the apparel condition, 9 kPa, appropriate to the 185.5 g/m² reference fabric.

1.2 Endpoint by thread breakdown

For woven fabrics the standard endpoint is the **breakdown point**: inspection stops the test at set rub intervals and the endpoint is the number of rubs at which **two warp or weft threads are completely broken** (holes just appearing in knits; a defined appearance change for nonwovens and pile fabrics). Inspection intervals widen as the test lengthens, because abrasion life is compared on a near-logarithmic scale — the difference between 5,000 and 6,000 rubs matters less than the difference between 500 and 1,500. The endpoint method answers the user's question directly: how long until the fabric wears through?

1.3 Mass-loss method

Alternatively — and always as supporting evidence — specimens are weighed before abrasion and after a fixed number of rubs, and the loss is expressed as a percentage of the original mass:

$$\text{Mass loss \%} = \text{mass lost} / \text{original specimen mass} \times 100$$

Mass loss measures the whole wearing process, including the long early phase in which fibres and finish are shed before any thread breaks. Two fabrics can share a breakdown endpoint yet lose mass at quite different rates; the mass-loss curve (loss plotted at increasing rub counts) separates them. Weighing must follow re-conditioning at 65% RH and 20 °C, since a freshly abraded, warm, dry specimen does not hold standard regain.

1.4 Pilling — the companion surface failure

The same Martindale stations, run at low pressure with the specimen rubbed against itself or a standard fabric, assess **pilling**: loose fibres migrate, tangle into pills anchored by strong fibres, and the surface is graded against photographic standards (grade 5 = no pilling, grade 1 = dense severe pilling) after a fixed rub count. Pilling is an appearance failure, not a strength failure — pills may form thousands of rubs before any thread weakens — so it is graded and reported separately, never merged into the abrasion endpoint.

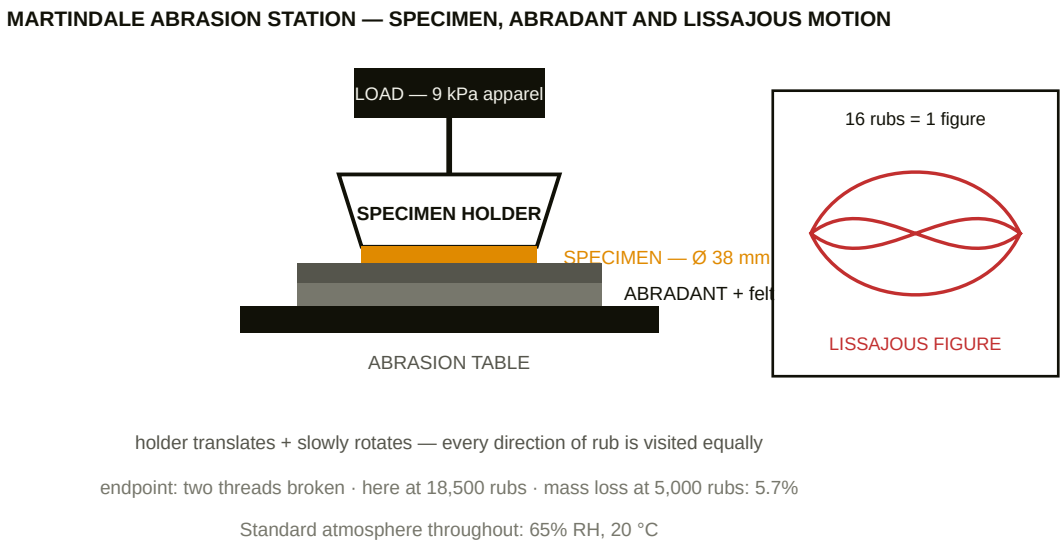


Figure 1 — One Martindale station: the loaded specimen holder (9 kPa for apparel) rubs the 38 mm specimen against the standard abradant on the table, driven through the Lissajous figure (inset) so abrasion arrives from all directions.

2. Apparatus and procedure

- **Apparatus:** Martindale abrasion tester with specimen holders and loading weights for 9 kPa; standard worsted abradant fabric and felt backing, renewed as specified; 38 mm specimen cutter;

balance reading to 0.001 g; magnifier for endpoint inspection; conditioned reference fabric.

- **Mounting:** cut 38 mm specimens from spread positions, mount each flat and wrinkle-free in its holder with the face outward; mount fresh abradant over felt on the table and confirm it is smooth and correctly tensioned.
- **Loading and running:** fit the apparel loading (9 kPa nominal pressure), set the counter to zero and start. Stop at the standard inspection intervals, examine each specimen under the magnifier without removing it from its holder where the instrument permits, and return it to the same station and orientation.
- **Breakdown endpoint:** record the rubs at which two threads are first completely broken in a specimen; test at least three specimens and report the endpoint behaviour (lowest station governs in most specifications).
- **Mass-loss check:** for the parallel specimens, condition and weigh before the test; after exactly 5,000 rubs, remove loose debris gently, re-condition at 65% RH and 20 °C, reweigh, and compute percentage mass loss from the specimen's original mass derived from its area and the fabric GSM.
- **Pilling note:** where required, run the pilling configuration separately and grade the surface against the photographic standards after the stated rub count; report the grade separately from abrasion results.

3. Worked calculation

Breakdown endpoint. Under 9 kPa apparel pressure against the standard abradant, inspection found the second thread broken at:

$$\text{Endpoint} = 18{,}500 \text{ rubs (two threads broken)}$$

For a 185.5 g/m² plain-weave cotton apparel fabric this is a solid working life — comfortably in the range expected of shirting and light suiting weights, and consistent with the tensile profile of Experiment 6: the threads that break at 420 N warp-way do not surrender quickly to surface wear either.

Mass-loss check at 5,000 rubs. The specimen is the standard 38 mm diameter disc:

$$\text{Specimen area} = \pi \times (1.9 \text{ cm})^2 = \pi \times 3.61 = 11.34 \text{ cm}^2$$

$$\text{Original mass} = 11.34 \text{ cm}^2 \times 185.5 \text{ g/m}^2 = 11.34 \times 0.01855 \text{ g} = 0.2098 \text{ g}$$

After 5,000 rubs and re-conditioning, the measured loss was 12 mg = 0.012 g:

$$\text{Mass loss \%} = 0.012 / 0.2098 \times 100 = 5.7\%$$

QUANTITY	VALUE

Breakdown endpoint (two threads broken, 9 kPa)	18,500 rubs
Specimen disc diameter / area	38 mm / 11.34 cm²
Original specimen mass (at 185.5 g/m²)	0.2098 g
Mass lost at 5,000 rubs	12 mg
Mass loss at 5,000 rubs	5.7%

READING THE TWO RESULTS TOGETHER

At 5,000 rubs — barely a quarter of the way to breakdown — the fabric has already shed 5.7% of its mass as fibre and finish debris, yet no thread has broken. That is the normal abrasion story: a long thinning phase, then rapid structural failure once surface yarns are cut through. Specifications that set both a minimum endpoint and a maximum early mass loss are guarding both halves of that story.

4. Precautions

- Renew the abradant and felt at the specified intervals and never mid-test on one station: a glazed, fibre-clogged abradant rubs more gently and inflates the endpoint.
- Mount specimens perfectly flat; a crease concentrates rubbing on its ridge and breaks threads there thousands of rubs early.
- Inspect without disturbing the specimen's station or orientation — a remounted specimen restarts its wear pattern and the endpoint is no longer one continuous history.
- Re-condition before every mass-loss weighing and brush off only loose debris; embedded abradant fibres and moisture drift both falsify milligram weighings.
- Use the pressure for the end use — 9 kPa apparel here; results at 12 kPa are substantially lower and not interchangeable.
- Keep specimens from separate fabric positions (Experiment 1 variation applies to wear too) and report the number of stations with the endpoint.

Practice (viva preparation)

1. Describe the motion of the Martindale specimen holder and explain why that motion is used.
2. What pressure is used for apparel fabrics in this test, and what is the breakdown endpoint for a woven fabric?
3. A specimen disc is 38 mm in diameter. Find its area in cm².

4. The fabric GSM is 185.5 g/m². Find the original mass of the disc in Question 3.
5. After 5,000 rubs the disc has lost 12 mg. Find the percentage mass loss.
6. Why is the specimen re-conditioned before the mass-loss weighing?
7. Distinguish abrasion breakdown from pilling as failure modes, and state how each is reported.

Answer key

1. The holder is driven in a Lissajous pattern — a continuously changing curve that slowly rotates, so the specimen is rubbed from every direction in turn — with a slow rotation superimposed. Real wear attacks from shifting directions; a fixed-direction rub would cut a groove aligned with one thread system and give a result depending on weave direction rather than on the fabric's true wear resistance. One complete Lissajous figure equals 16 rubs.
2. 9 kPa nominal pressure for apparel fabrics (12 kPa is used for workwear and upholstery-type uses). The woven-fabric endpoint is the breakdown point: the number of rubs at which two threads — warp or weft — are completely broken, found by inspection at widening intervals. Here it was 18,500 rubs.
3. Radius = 1.9 cm. Area = $\pi \times 1.9^2 = \pi \times 3.61 = 11.34$ cm².
4. Mass = area × GSM = 11.34 cm² × 185.5 g/m². Converting, 185.5 g/m² = 0.01855 g/cm², so mass = $11.34 \times 0.01855 = 0.2098$ g.
5. Mass loss % = 0.012 g / 0.2098 g × 100 = 5.7%.
6. Abraded specimens come off the machine warm, dry and carrying disturbed regain, and milligram-level weighing is being compared against a GSM-derived mass that assumes standard regain. Re-conditioning at 65% RH and 20 °C restores the standard moisture state so the loss recorded is fibre and finish, not water.
7. Abrasion breakdown is a structural failure — threads wear through until two are broken — reported as an endpoint in rubs (with mass loss as supporting evidence). Pilling is an appearance failure — surface fibres tangle into anchored balls while the fabric remains structurally sound — reported as a grade from 5 (none) to 1 (severe) against photographic standards after a stated number of rubs. They are run in different Martindale configurations and never combined into one number.
